



AI-POWERED NETWORK ATTACK DETECTION SYSTEM AND ANALYST DASHBOARD

Göktaş VARAN, Mehmet Taha YILMAZ, Prof. Dr. Özgür Koray ŞAHİNGÖZ

Department of Computer Engineering, Biruni University, Istanbul, Turkey | 2025-2026

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INTRODUCTION

Cyber attacks are becoming more complex and frequent. Traditional rule-based intrusion detection systems may fail against new attack patterns. This project proposes an AI-powered intrusion detection system that classifies network traffic and visualizes alerts through an analyst dashboard.



PROBLEM

- Evolving attacks
- Imbalanced data
- Limited analyst visibility



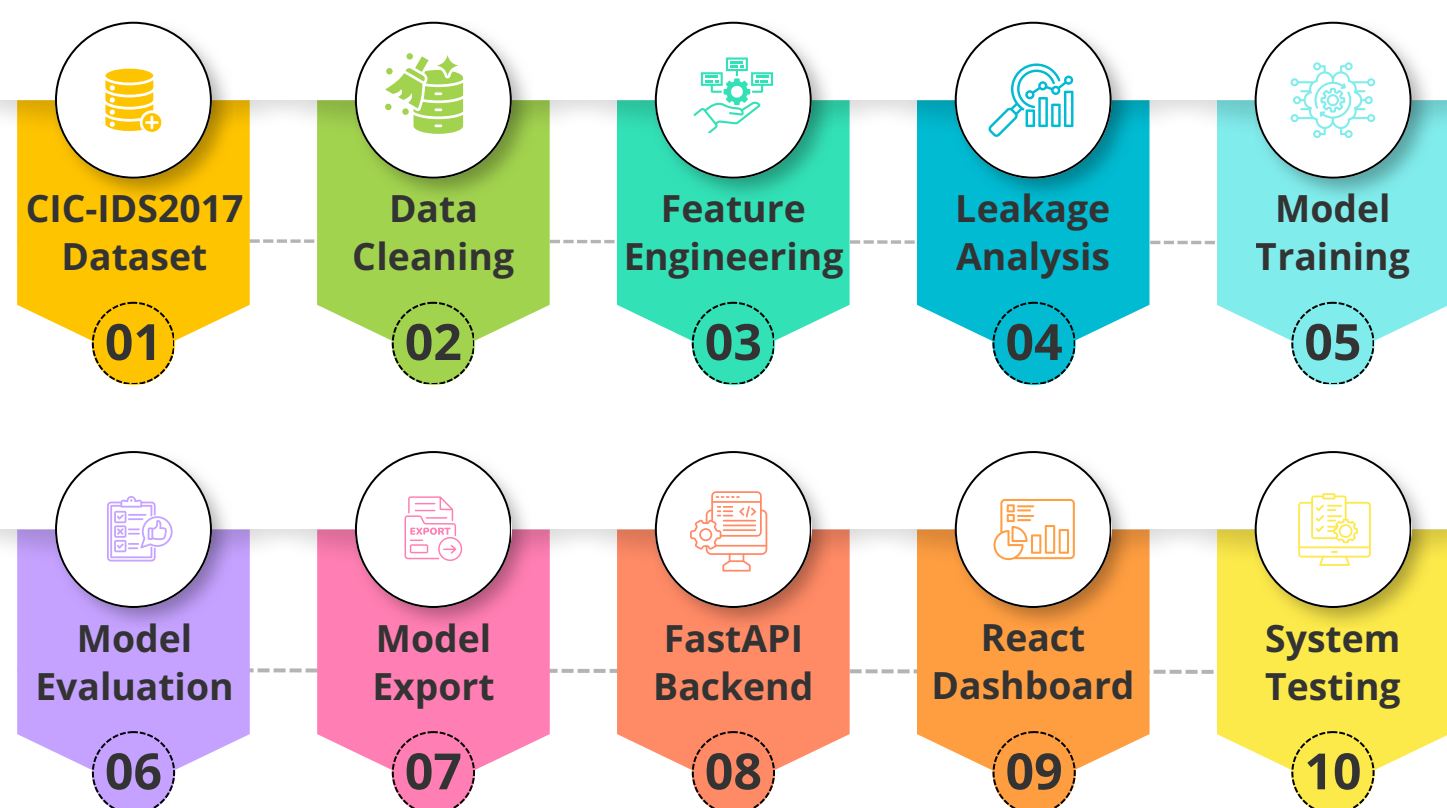
PROPOSED SOLUTION

- ML-based detection
- Binary & multiclass detection
- Dashboard with XAI



METHODOLOGY

The project follows an end-to-end machine learning pipeline from CIC-IDS2017 data preparation to dashboard integration. The trained models are exported and served through a FastAPI backend for monitoring.



Dataset

CIC-IDS2017
Network (Benign + Attack)
Traffic Flows

Models

Random Forest
Binary and Multiclass
Classification

Deployment

FastAPI + React
Analyst Dashboard



RESULTS

Model	Binary Acc.	Multiclass Acc.	Selection
RF	%99.85	%99.69	Final
XGBoost	%99.82	%99.73	Compared
CatBoost	%99.82	%96.29	Compared

1 Binary Classification

Random Forest

Accuracy

Per-file
%99.85

Macro F1

Per-file
%99.80



Per-file evaluation was selected as the primary benchmark to ensure balanced class representation across all attack categories, providing a controlled and reproducible assessment of model performance.

2 Multiclass Classification

Random Forest

Accuracy

Per-file
%99.69

Weighted F1

Per-file
%99.69

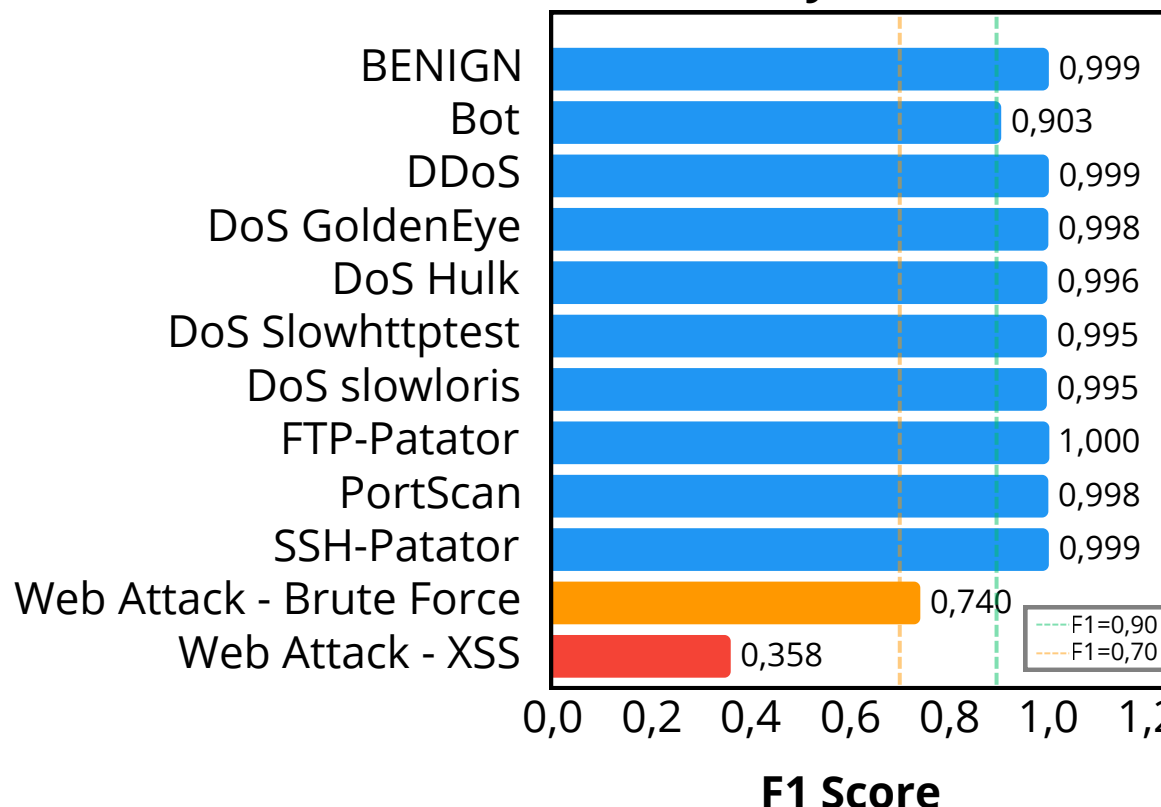
Macro F1

Per-file
%91.87



Macro F1 (91.87%) reflects reduced performance on minority web attack classes despite high overall accuracy.

Per-Class F1 Score Analysis



FedAvg Federated Learning Results

Metric	Value
Accuracy	%96.95
Weighted F1	%96.72
Macro F1	%74.51
Communication Rounds	5
Clients	5
Local Epochs	2



Most major attack categories achieved high F1-scores, while Web Attack-XSS remained difficult due to limited and imbalanced samples.



Key Observations

- Random Forest was selected for final deployment due to strong macro F1, stability, and integration simplicity.
- Per-file evaluation demonstrated strong binary and multiclass performance across all major attack categories.
- Minority web attacks, especially XSS, remained the most difficult classes.
- Leakage-sensitive and unstable features were removed to reduce shortcut learning.



DASHBOARD

The dashboard processes each network flow through a two-stage ML pipeline. Detected attacks are automatically classified by type and explained using per-sample SHAP feature contributions.

Input Record	Binary RF	Multiclass RF	Per-Sample XAI	Alert Generated
FlowID: #4821 Timestamp: 16:03 Avg Pkt Size: 7.5 Duration: 0.03s Fwd Pkts/s: 1,186 Total Len Fwd: 24	Prediction: ATTACK Confidence: %99.8	Attack Type: DDoS Confidence: %97.1	Average Pkt Size act_data_pkt_fwd Total Length Fwd	Severity: CRITICAL Attack Type: DDoS Top Feature: AvgPkt SHAP Value: +0.082 Sent to: Analyst Dashboard



CONCLUSION

- An end-to-end IDS prototype was developed by combining ML models, FastAPI backend, React dashboard, and SHAP-based explanations.
- Random Forest achieved strong per-file binary and multiclass performance.
- Per-file evaluation confirmed strong generalization across all major attack categories.
- Minority web attacks require further data enrichment and class-specific improvement.
- Future work includes live packet capture, persistent storage, SIEM integration, and model drift monitoring.



GitHub
Repository

Full Report Thesis & Article and
References Included